

Alexander Kai Chen

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Mission

Intersecting computer science and classics to explore how language encodes thought. Building explainable systems to make automation and reasoning verifiable, transparent, and accessible.

Publications

2025 — iOS as Acceleration

NeurIPS 2025 International Conference, Efficient Reasoning Workshop

- Presented novel distributed computing framework to harness ubiquitous, underutilized handheld phones as local compute accelerators
- Paper (arxiv.org/abs/2512.22180) published in NeurIPS 2025 ER Workshop

SWE & AI Experience

Summer 2025, Summer 2026 — Software & Data Engineer

State Farm Enterprise Technology Division, Intern

- Built production and internal high-volume data processing applications
- AWS, Azure, Java, JavaScript, SCRUM Agile, agentic AI workflows

Summer 2024, 2026 — Logic & Explainable AI Research

UTD Applied Logic & Programming Systems Lab (ALPS), Intern

- XAI education research, explored Prolog and logic programming with Professor Gopal Gupta
- Answer Set Programming (ASP), Prolog, ML

2022–2025 — FRC Robotics Lead Analytics Developer

FIRST Robotics, Team 2714 BBQ

- Developed data scouting software for FRC team, deployed to world championships in 2024 and 2025.
- Android Applications, Web Interfaces, Java, JavaScript

Engineering & Research Skills

- **Programming Languages:** Python, Java, Lean4, Prolog, C++, Lisp (Clojure), Swift, JavaScript
- **AI/ML:** Neuralsymbolic Systems, Explainable AI, Distributed Computing & Parallelism, Agentic LLMs
- **Theory:** Algorithms & Data Structures, Software Development, Formal Verification, Discrete Mathematics
- **Frameworks & Tools:** AWS, Azure, Angular, FastAPI, PyTorch, Mathlib
- **Workflows:** SCRUM, Agile, Agentic Orchestration

Education

2023–Present — Highschool

Homeschool

- 4.0 GPA, 1600 SAT, 6x APs score 5, currently completing Junior year

2025–Present — Dual Credit College Courses

Collin College

- Discrete Math (Grade: A, 100%)
- Linear Algebra (in-progress)

2023–Present — Online College Courses

Online

- “Data Structures & Algorithms Specialization” at UCSD
- “Build a Modern Computer from First Principles” at HUJI

2023–Present — C.S. & A.I.

Self-Directed Study, Mentored

- Neuralsymbolic AI, XAI, Distributed Learning, NLP, Agentic LRMs

2023–Present — Classical Languages & Philosophy

Highschool Classical Education, Self-Directed Study

- Studying classical, ecclesiastical Latin, Greek & Roman philosophy, debate rhetoric & argumentation

Honors & Achievements

2025–2026 — ISEF State Fair

Texas Science & Engineering Fair

- Placed 2nd in Software Design category, earning full scholarship to Governor’s Science & Technology Champion’s Academy summer research program

2025–2026 — ISEF Regional Fair

Dallas Regional Science & Engineering Fair

- Placed 1st winning Software Design category with “MobilePipe: iPhones as Parallel Compute Accelerators for Local Machine Learning,” advanced to TXSEF

2022–2026 — Speech & Debate

Coolidge Foundation Debate (25–26), NCFCA Speech & Debate (22–25)

- Qualified for national Coolidge Cup; consistently advanced to elimination rounds in team policy debate & persuasive platform, apologetics speaking.
- Demonstrated professional communication, applied reasoning, and public presentation skills

2023–2025 — USACO

USA Computing Olympiad, Competitive Algorithmic Programming

- Promoted to Gold Division in 2025; previously promoted to Silver Division in 2023.

2022–2025 — FRC Robotics World Championships

FIRST Robotics Competition

- Served robotics programming team, led scouting data analytics application development, advanced to world finals championship 2024, 2025